(3) $(cf,g)=\sum_{i,j}\frac{ca_ib_j}{i+j+1}=c\sum_{i,j}\frac{a_ib_j}{i+j+1}=c(f,g)$

(4) $(f,f)=\sum_{i,j}\frac{a_ia_j}{i+j+1}=\int_0^1 f^2(x)dx\geq 0$

等号成立当且仅当 $f(x)=0$

∴ (f,g) 为内积

8.(1) ① $(AB,A)=tr(BA')=tr((BA')')=tr(AB')=(A,B)$

② $(A+B,C)=tr((A+B)C')=tr(AC'+BC')=tr(AC')+tr(BC')=(A,C)+(B,C)$

③ $(cA,B)=tr(cAB')=c\,tr(AB')=c(A,B)$

④ $(A,A)=tr(A^2)\geq 0$ 当且仅当 $A=0$

(2) 同上

9.2.1 (1) 令 $u_1=v_1=(1,0,0)$

∴ $u_2=(1,1,0)-\frac{1}{1^2}(1,0,0)=(0,1,0)$

$u_3=(1,1,1)-\frac{1}{1}(0,1,0)-\frac{1}{1}(1,0,0)=(0,0,1)$ ∴ $\{u_1,u_2,u_3\}$ 为 V 的组成基

(2) $v_1=(1,2,2,-1)$

$v_2=(1,1,-5,3)-\frac{-10}{10}(1,2,2,-1)=(2,3,-3,2)$

$v_3=(3,2,8,-7)-\frac{30}{10}(1,2,2,-1)+\frac{26}{26}(2,3,-3,2)=(2,-1,-1,-2)$

再取 $w_1=\frac{1}{\sqrt{10}}v_1=\frac{1}{\sqrt{10}}(1,2,2,-1)$ $w_2=\frac{1}{\sqrt{26}}(2,3,-3,2)$ $w_3=\frac{1}{10}(2,-1,-1,-2)$

则 $\{w_1,w_2,w_3\}$ 为 V 的一组标准正交基

(3) $v_1=(1,1,-1,-2)$

$v_2=(5,8,-2,-3)-3(1,1,-1,-2)=(2,5,1,3)$

$v_3=(3,9,3,8)+(1,1,-1,-2)+2(2,5,1,3)=(0,0,0,0)$

令 $w_1=\frac{1}{\sqrt{7}}(1,1,-1,-2)$ $w_2=\frac{1}{\sqrt{39}}(2,5,1,3)$

∴ $\{w_1,w_2\}$ 是一组正交基

2. 由9.1 有 $\alpha=(\alpha,e_1)e_1+\cdots+(\alpha,e_n)e_n$

$\beta=(\beta,e_1)e_1+\cdots+(\beta,e_n)e_n$

∴ $(\alpha,\beta)=(\alpha,e_1)e_1+\cdots+(\alpha,e_n)e_n,(\beta,e_1)e_1+\cdots+(\beta,e_n)e_n)$

$=(\alpha,e_1)\overline{(\beta,e_1)}+\cdots+(\alpha,e_n)\overline{(\beta,e_n)}$

3. 设 $V_2=\{v\in V|(v,u_i)=0,i=1,2,\cdots m\}$

∴ 由 $U^\perp$ 定义 $U^\perp=\{v\in V|(v,U)=0\}$ ∴ $U^\perp\subset V_2$

而对 $\forall w\in V_2$，及 $\forall u\in U$ 有 $(u,w)=k_1(u_1,w)+\cdots+k_n(u_n,w)=0$

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∴ $W \perp U$ ∴ $U_2 \subseteq U^\perp$ ∴ $U_2 = U^\perp$

4. (1) ∵ $V = U \oplus U^\perp = U \oplus U^\perp \oplus (U^\perp)^\perp$ ∴ $U = (U^\perp)^\perp = V/U$

(2) 显然 $(U_1+U_2)^\perp \subset U_1^\perp$ $(U_1+U_2)^\perp \subset U_2^\perp$ ∴ $(U_1+U_2)^\perp \subset U_1^\perp \cap U_2^\perp$

任取 $\alpha \in U_1^\perp \cap U_2^\perp$ $\beta \in U_1+U_2$ 设 $\beta = \beta_1+\beta_2$ $(\beta_1 \in U_1, \beta_2 \in U_2)$

∴ $(\alpha, \beta) = (\alpha, \beta_1) + (\alpha, \beta_2) = 0$

∴ $\alpha \perp (U_1+U_2)$ ∴ $\alpha \in (U_1+U_2)^\perp$ ∴ $U_1^\perp \cap U_2^\perp \subset (U_1+U_2)^\perp$

∴ $U_1^\perp \cap U_2^\perp = (U_1+U_2)^\perp$

5. (3) ~~$U_1 \cap U_2$~~ 任取 ~~$(U_1 \cap U_2)^\perp$~~ =

由(1),(2) $(U_1^\perp + U_2^\perp)^\perp = U_1 \cap U_2$

∴ $U_1^\perp + U_2^\perp = (U_1 \cap U_2)^\perp$

(4) ~~$\forall v \in S$~~ 若 $v \in V^\perp$ 则 $(v,v) = 0$ ∴ $v = 0$ ∴ $V^\perp = 0$

对 0, $\forall v \in V$, 有 $(v, 0) = 0$. ∴ $0^\perp = V$

5. (1) 若 $\alpha \in S^\perp$, $\beta \in S^\perp$, $c \neq 0$ 则 $(\alpha+\beta, S) = (\alpha, S) + (\beta, S) = 0$

$(c\alpha, S) = c(\alpha, S) = 0$

∴ $\alpha+\beta$, $c\alpha \in S^\perp$

∴ $S^\perp$ 是 V 的子空间

(2) 设由 S 生成的子空间为 U, 则 $U^\perp \subset S^\perp$, 而对 $\forall v \in S^\perp$, $u \in U$, 将 u 拆为 S 中元素, 即 $u = k_1 s_1 + k_2 s_2 + \cdots + k_n s_n$ ∴ $(u, v) = (\sum_{i=1}^{n} k_i s_i, v) = 0$

∴ $v \in U^\perp$ ∴ $S^\perp \subset U^\perp$ ∴ $S^\perp = U^\perp$

∴ $S = (S^\perp)^\perp = (U^\perp)^\perp = V$

6. 解得 解空为 $(x_1, x_2, x_3, x_4)^T = k_1(0,1,0,1) + k_2(-6,9,1,0)^T$

∴ $U^\perp$ 适合的方程为 $\begin{pmatrix} 0 & 1 & 0 & 1 \\ -6 & 9 & 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = 0$

7. 设 $A \in V_1$, $B \in V_2$

∴ $(A, B) = tr(AB') = -tr(AB) = -tr(BA) = -tr(BA^T) = -tr(B, A) = -(A, B)$

∴ $(A, B) = 0$ ∴ ~~$V = V_1$~~

又∵ $V = V_1 \oplus V_2$ ∴ $V = V_1 \perp \oplus V_2$

8. 设 $\alpha = x_1 e_1 + \cdots + x_n e_n$ 若 α 满足 $(\alpha, e_i) = c_i$, 则

∴ $\begin{cases} (e_1, e_1)x_1 + (e_1, e_2)x_2 + \cdots + (e_1, e_n)x_n = c_1 \\ \cdots\cdots \\ (e_1, e_n)x_1 + (e_2, e_n)x_2 + \cdots + (e_n, e_n)x_n = c_n \end{cases}$

∴ 得到 n 元线性方程组, 其系数矩阵为 Gram 阵, 为正定阵 ∴ 该方程有唯一解.

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即α存在且唯一

9、由Gram-Schmidt, $(u_1, u_2 \cdots u_n) = (v_1, v_2 \cdots v_n)A$

A为对角元为1的上三角矩阵, $A=(a_{ij})$

对 $G(u_1 \cdots u_n)$的 (k,l)位元素

$(u_k, u_l) = (\sum_{i=1}^{n} a_{ik} v_i, \sum_{j=1}^{n} a_{jl} v_j) = \sum_{ij} a_{ik} a_{jl} (v_i, v_j) = \sum_{i} a_{ik} a_{il}$

即 $A'G(v_1 \cdots v_n)A$的(k,l)位元素

$\therefore G(u_1 \cdots u_n) = A'G(v_1 \cdots v_n)A$

$\therefore |G(u_1 \cdots u_n)| = |A|^2 |G(v_1 \cdots v_n)| = |G(v_1 \cdots v_n)| = \|v_1\|^2 \cdots \|v_n\|^2$

10、必要性由9可得

充分性：反证，若 $u_1 \cdots u_m$线性相关即 $\exists i, \exists u_i$

设 $u_i = k_1 u_1 + k_2 v_2 + \cdots + k_{i-1} v_{i-1} + k_{i+1} v_{i+1} + \cdots + k_m v_m$

又设 $G(u_1 \cdots u_m) = (\alpha_1, \alpha_2 \cdots \alpha_m)$

则 $\alpha_i = k_1 \alpha_1 + k_2 \alpha + \cdots + k_m \alpha_m$

$\therefore |G(u_1 \cdots u_m)| = 0$

得证

11、由 $\alpha' G \alpha \geq 0$ 得 $|G(u_1 \cdots u_m)| \geq 0$

对右侧：(1) $|G(u_1 \cdots u_m)| = 0$，显然成立，此时若等号成立则 $\exists \|u_i\| = 0$

(2) $|G(u_1 \cdots u_m)| > 0$ 则 $G(u_1 \cdots u_m)$正定

由9, $|G(u_1 \cdots u_m)| = |G(v_1 \cdots v_m)| = \|v_1\|^2 \cdots \|v_m\|^2$ 其中V是正交向量组

由 Gram-Schmidt 公式 $v_{k+1} = u_{k+1} - \sum_{j=1}^{k} \frac{(u_{k+1}, v_j)}{\|v_j\|^2} v_j$

进而由 $v_i \perp v_j$ $\|u_{k+1}\|^2 = \|v_{k+1}\|^2 + \sum_{j=1}^{k} \frac{(u_{k+1}, v_j)^2}{\|v_j\|^2} \geq \|v_{k+1}\|^2$

$\therefore |G(u_1 \cdots u_m)| \leq \|u_1\|^2 \cdots \|u_m\|^2$

等号成立当且仅当 $v_i = v_i \ (i=1, 2 \cdots m)$ 即 u_i两两正交

12、设$A=(u_1 u_2 \cdots u_n)$ 则 $|A^2| = |A'A| = |G(u_1 \cdots u_n)| \leq \prod_{j=1}^{n} \|u_j\|^2 = \prod_{j=1}^{n} \sum_{i=1}^{n} a_{ij}^2 \leq \prod_{j=1}^{n} \sum_{i=1}^{n} \|u_i\|^2$

13、~~当 $j \leq k$, $(u_k(x), x^j) = \int_{-1}^{1} \frac{d^k}{dx^k}[(x^2-1)^k] x^j dx = -j \int_{-1}^{1} \frac{d^{k-1}}{dx^{k-1}}[(x^2-1)^k] x^{j-1} dx$~~

~~反复操作可得 $j<k$时 $(u_k(x), x^j) = 0$ $(u_k(x), x^k) = (-1)^k k! \int_{-1}^{1} (x^2-1) dx$~~

~~对 $u(x)$, $\because (x^2-1)^k$ 首项为 x^{2k} $\therefore u(x)$首~~ 对 $P_0(x)$, 显然 $(P_0(x), P_i(x)) = 0 \ (i \neq 0)$

~~设 u~~ 显然 $\{1, x, x^2, \cdots x^n\}$是 V的一组基 $\therefore$ V的维数为 $n+1$

设 $u_k(x) = \frac{d^k}{dx^k}[(x^2-1)^k]$，若 $j<k$，有 $\frac{d^j}{dx^j}[(x^2-1)^k]|_{x=\pm 1} = 0$ $\therefore$ 分部积分有

$\therefore (u_k(x), x^j) = \int_{-1}^{1} \frac{d^k}{dx^k}[(x^2-1)^k] x^j dx = -j \int_{-1}^{1} \frac{d^{k-1}}{dx^{k-1}}[(x^2-1)^k] x^{j-1} dx$

重复可得 $(u_k(x), x^j) = 0$ 而 $u_k(x)$为 k次多项式

$\therefore (u_k(x), u_j(x)) = 0 \ (k \neq j)$ 进而 $(P_k(x), P_j(x)) = 0 \ (k \neq j$，且$k \neq 0, j \neq 0)$

$\therefore P_k(x)$间相互正交 又 $\because \{P_0(x) \cdots P_k(x)\}$有$(n+1)$个 $\therefore P_k(x)$构成 V的一组且正交基

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1. φ 对应 $A=\begin{pmatrix}1 & 1\\ -2 & -1\end{pmatrix}$ φ^* 对应 $\bar{A}'=\begin{pmatrix}1 & -2\\ -1 & -1\end{pmatrix}$

$\therefore \varphi^*(x)=(x_1-2x_2, -x_1-x_2)$

2. 设 $\alpha\in\ker\varphi$ $\beta^*\in \mathrm{Im}\varphi^*$ 则 $\beta^*=\varphi^*(\beta)$

$\therefore (\alpha,\beta^*)=(\alpha,\varphi^*(\beta))=(\varphi(\alpha),\beta)=(0,\beta)=0$

$\therefore \ker\varphi\subseteq(\mathrm{Im}\varphi^*)^\perp$

设 $\alpha\in(\mathrm{Im}\varphi^*)^\perp$ $\beta^*\in\mathrm{Im}\varphi^*$, $\beta^*=\varphi^*(\beta)$

$\therefore (\alpha,\beta)=0=(\alpha,\varphi^*(\beta))=(\varphi(\alpha),\beta)$

令 $\beta=\varphi(\alpha)$ $\therefore (\varphi(\alpha),\varphi(\alpha))=0$ $\therefore \varphi(\alpha)=0$

$\therefore \alpha\in\ker\varphi$ $\therefore \ker\varphi\supseteq(\mathrm{Im}\varphi^*)^\perp$

$\therefore \ker\varphi=(\mathrm{Im}\varphi^*)^\perp$ $\therefore (\ker\varphi)^\perp=\mathrm{Im}\varphi^*$

3. $\because\varphi$ 是线性自同构 $\therefore\varphi$ 对应的矩阵 A 为可逆阵

$\therefore\varphi^*$ 对应的 $\bar{A}'$ 可逆 $\therefore\varphi^*$ 为线性自同构

设 φ 对应矩阵为 A 则 $(\varphi^*)^{-1}$ 对应 $(\bar{A}')^{-1}$ $(\varphi^{-1})^*$ 对应 $(\overline{A^{-1}})'=(\bar{A}')^{-1}$

$\therefore(\varphi^*)^{-1}=(\varphi^{-1})^*$

4. $\varphi(x+y)=(x+y,\alpha)\beta=(x,\alpha)\beta+(y,\alpha)\beta=\varphi(x)+\varphi(y)$

$\varphi(cx)=(cx,\alpha)\beta=c(x,\alpha)\beta=c\varphi(x)$

$\therefore\varphi$ 是 V 上的线性变化

令 $(\varphi(x),y)=(k\alpha,\beta,y)=(x,\alpha)(\beta,y)=(x,\alpha(\beta,y))=(x,\varphi^*(y))$

$\therefore\varphi^*(x)=k(x,\beta)\alpha$

φ 对应 E_{21} φ^* 对应 E_{12}

5. 对 B, $(\varphi(A),B)=(TA,B)=\mathrm{tr}(TAB')=\mathrm{tr}(AB'T)=\mathrm{tr}(A(\overline{TB})')=(A,\overline{TB})$

$\therefore\varphi^*(A)=\overline{T}A$

6. $(\varphi(x),x)=(\lambda_1x,x)=\bar{\lambda}_1(x,\varphi^*(x))=(x,\lambda_2x)$

$\therefore(\lambda_1-\bar{\lambda}_2)(x,x)=0$ $\therefore x\neq 0$ 或 $\lambda_1=\bar{\lambda}_2$

7. 设 φ 对应矩阵为 A 设 $g(A)=a_nA^n+a_{n-1}A^{n-1}+\cdots+a_1A+a_0=0$

$\therefore\bar{g}(\bar{A}')=\bar{a}_n(\bar{A}')^n+\cdots+\bar{a}_1(\bar{A}')+\bar{a}_0$

$=\overline{[a_nA^n+\cdots+a_1A+a_0]}'=\overline{g(A)}'=0$

设 φ^* 对应的极小多项式为 $m(x)$, 则 $m(x)\mid\bar{g}(x)$,

同上可得 $g(x)\mid\bar{m}(x)$ $\therefore m(x)=g(x)$

8. $\forall u,v\in U$ $(\varphi|_U(u),v)=(\varphi(u),v)=(u,\varphi^*(v))=(u,\varphi^*|_U(v))$

$\therefore(\varphi|_U)^*=\varphi^*|_U$

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